

Load Interconnection Request System Impact Study Report

For Queue Project Q22-2021

Blackhawk 138 kV
May 02, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2021. The Interconnection Customer proposes a 100.9 MW large load (data center) facility interconnecting at the Blackhawk 138 kV substation in the Virginia area. The request was received on March 01, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 440 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$328,000.

3. Network Impacts

3.1 Load Deliverability Analysis

1. Facility: Kean Mtn And Grassy Crk Substation / Clinch River 345 KV
Contingency: Kean Mtn And Grassy Creek 345 kV/1 kV XFMR
Loading: 102.5% (Rating: 1,254 MVA)
DFAX: 8.7%
Pre-project loading: 101.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Clinch River / Kean Mtn And Grassy Crk 345 KV	Reconductor line	\$26,897,000	1% (\$328,000)
	Total allocated to project		\$328,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.